ChatGPT

Executive Summary

Polymetallic (ferromanganese) nodules – potato-sized metal-rich concretions on deep-ocean plains – are world-wide (Pacific, Indian and lesser Atlantic) and contain economically significant **copper** and **aluminium** among other metals. Nodule growth is extremely slow (few mm/Myr) via concentric layering of Fe–Mn oxides and silicates around a nucleus【63†L216-L224】【62†L25-L31】. Typical nodules contain ~29% Mn, ~6% Fe, ~5% SiO_2 and trace metals (globally ~1.4% Ni, 0.25% Co, 1–1.4% Cu, 3% Al【62†L25-L31】【32†L100-L104】). Nodule fields occur in abyssal basins at 3.5–6 km depth with very low sedimentation (Antarctic Bottom Water flows keep sediments thin【63†L213-L221】【92†L262-L270】). The largest field is the **Clarion–Clipperton Zone (CCZ)** in the Pacific (≈9×10^6 km²), with nodules concentrated in smooth abyssal plains. Other major provinces include the **Peru Basin** (Pacific), **Penrhyn Basin** (Cook Islands region) and the **Central Indian Basin**【92†L288-L297】【92†L307-L316】. Global CCZ tonnage is enormous: an ISA estimate suggests ~27×10^9 t nodules (wet) in the CCZ, containing ~340 Mt Ni and 290 Mt Cu【46†L1175-L1178】 (i.e. ~1.3% Ni, 1.1% Cu by mass). World Ocean Review data indicate 21×10^9 t in the CCZ【92†L293-L300】, and additional tens of billions more in Peru/Indian fields. Typical resource models use ~4 kg/m² (wet) cutoff, equating to ~1–2% combined Ni+Cu+Co【63†L222-L224】【43†L1119-L1127】.

Deep-sea nodule exploration uses systematic mapping and sampling. Multibeam echosounders and acoustic backscatter surveys delineate nodule-rich areas【45†L1189-L1197】, with box cores and grabs (5–30 km spacing) for ground-truth【43†L1133-L1138】【45†L1196-L1200】. Industry studies (e.g. Tonga Offshore Mining's TOML) report inferred resources from ~75,000 km² license areas, using ordinary kriging on historic grab/core data【43†L1119-L1127】【22†L1130-L1138】. TOML's inferred resource (2012) covers 440 Mt nodules (wet) at ~1.2% Ni, 0.24% Co, 1.1% Cu【43†L1119-L1127】. China, India and Germany similarly sample grids and classify "prospective nodule fields" by nodule size/abundance【45†L1189-L1197】.

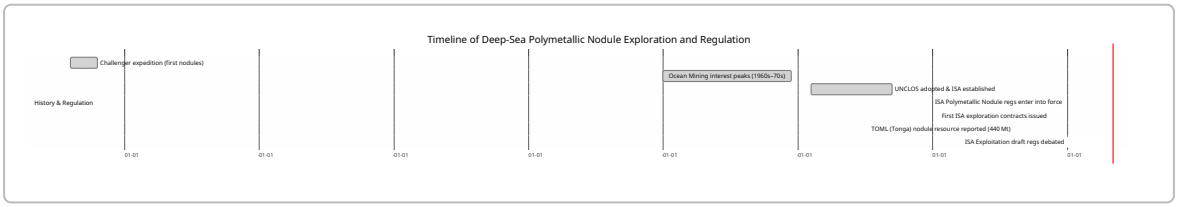
Legally, nodules in international waters fall under UNCLOS "common heritage" rules administered by the ISA. The 2000 ISA Mining Regulations govern exploration; a draft exploitation code (requiring environmental studies, benefit-sharing, insurance, etc.) is under negotiation. ISA has issued ~15-year exploration contracts (renewable) to countries and entities (e.g. Nauru, China, Russia, India, Germany, Tonga, Belgium, UK, etc.) covering ~75,000 km² each【63†L232-L239】【43†L1119-L1127】. (The CCZ contains all but one of these, the other being in the Central Indian Basin【63†L232-L239】.) Any commercial mining will require strict environmental management and compliance with UNCLOS articles (e.g. art.137–145, EIA obligations).

Mining technology concepts draw on trials from the 1970s–80s and recent prototypes. Collector vehicles (crawler/seafloor tractors) sweeping the seafloor ("like a potato harvester"【92†L281-L286】) are proposed. These machines would vacuum or scoop nodules and pump a slurry up an umbilical to a support vessel. For example, a dual-jet suction device has been modeled to achieve >80% pick-up efficiency for large nodules【61†L174-L181】. Modern trials include lander tests (e.g. by Global Sea Mineral Resources) and the recent Nauru ORP (Observation/Research Platform) campaign. Cost-viability depends on metal prices: nodules typically require *cutoff* ~4 kg/m² (wet) to be economic【63†L222-L224】【43†L1119-L1127】, or combined base-metal grade ≥2%. With Ni, Cu prices in the range US$10–20k/t, only the richest fields (high nodule density and metal content) are likely viable. All known

deposits have grades comparable to or slightly below rich terrestrial ores, so long-term metal demand (especially Ni/Cu for batteries) is the main driver.

Environmental risks are a key concern. Mining would obliterate the hard-substrate nodule habitat and create sediment plumes that can smother fauna. Deep-sea ecosystems are slow to recover and harbor unique biodiversity. Studies emphasize **precautionary management** and robust baseline data before any mining【6†L47-L54】【6†L79-L83】. The ISA and researchers advocate no-go zones (e.g. ISA-designated APEIs), adaptive management and ongoing monitoring. While deep-ocean mining might avoid some terrestrial impacts (e.g. no tailings dam), it raises novel risks. Future mitigation could include improved collector designs (low-turbidity pumps), buffer zones between mining blocks and detailed sediment plume modelling, but all proposals remain untested at scale.

Conclusions: The CCZ and certain Pacific/Indian basins host the world's best copper-bearing nodules. A typical CCZ license (like TOML) yields ~1.1% Cu (±0.5% variability), while aluminium (~3%) is essentially constant (fixed by silicate matrix【62†L25-L31】). Peru Basin nodules have similar metal ratios but generally lower densities (≈10 kg/m^2 vs 15–75 kg/m^2 in CCZ【92†L293-L300】). Resource estimates remain preliminary (many inferred only) and rest on sparse sampling. High uncertainty gaps include detailed Al variation, precise deep-sea ecosystem responses, and future metal prices. Nevertheless, geology strongly favors CCZ and comparable Pacific basins for near-term exploitation. Continued studies (including sampling and imaging campaigns) and finalized ISA regulations will determine which fields are truly feasible.




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flowchart LR
    A[Nucleus (shark tooth, pumice, sediment fragment)] --> B[Hydrogenetic
accretion<br/>(precipitation from seawater)]
    A --> C[Diagenetic accretion<br/>(precipitation from sediment porewater)]
    B --> D[Nodule growth<br/>(layered Fe–Mn oxide)]
    C --> D
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Table: Key Polymetallic Nodule Fields (Pacific & Indian Oceans). Values are approximate. Sources: ISA/industry reports and surveys【92†L293-L300】【43†L1119-L1127】【46†L1175-L1178】.

Field	Location & Depth	Cu (wt%)	Al (wt%)	Resource (wet nodules)	Accessibility / Relief	Legal Status	Note Envir
Clarion–Clipperton Zone (CCZ)	Pacific Ocean (≃ 8°–20°N, 120°–160°W); 4–6 km	≈1.0–1.2%【43†L1119-L1127】	~3%【62†L25-L31】	~27×10^9 t 【46†L1175-L1178】	Mostly flat abyssal hills (<20% slopes for ~98% of area 【43†L1111-L1118】)	ABNJ; multiple ISA contracts (e.g. Nauru, China, Germany, Tonga)	High nodu dens 75 kg 【92† L300 rich f diver preca APEIs desig
Peru Basin	Pacific (≈4°–10°S, 90°–110°W); 4–5 km	≈0.8–1.0% (estimated)	~3% (assumed)	~~40–50×10^9 t (rough calc.)~	Broad abyssal plain (low relief)	ABNJ; no active contracts reported	Mode nodu dens (≈10 【92† L306 fewe sea s than
Penrhyn Basin (Cook Is.)	South Pacific, near Cook Is.; ~4–5 km	≈1.0% (typical Pacific)	~3%	~3–5×10^9 t (est.)	Possibly gentle basin; some rise topography	Likely Cook Islands' EEZ (no ISA contract)	Smal high- field; unde natio juris
Central Indian Ocean	Indian Ocean (≈12°S, 85°E); ~4.8 km	≈0.9% (similar to CCZ)	~3%	~~4×10^9 t (field)~~ 【92†L313-L316】	Abyssal plain (5 kg/m² average 【92†L313-L316】)	ABNJ; ISA contracts (India, China, Korea)	Sole India field; some lowe abun than
Other (e.g. Atlantic)	e.g. Angola Basin, etc.; depths ~4–5 km	Data sparse	~3%?	Unknown	Often steeper slopes or sediment veneer	None (Atlantic nodules not licenced by ISA)	Smal patch foun less e

Data Gaps & Assumptions: Many resource figures are *inferred* from limited sampling (mostly box cores and grabs)【43†L1133-L1140】【22†L1130-L1138】. *Aluminium* content is not regularly reported in field studies; we assume ~3% Al (consistent with global average【62†L25-L31】) for all nodules. The above tonnages are based on generous area and density assumptions; true *recoverable* metal amounts will be lower after loss and processing. Cutoff criteria (nodule mass or metal grade) and future metal prices will greatly affect viability. Environmental sensitivity is an emerging assessment; our table qualitatively

notes that the richest areas (e.g. CCZ) also coincide with high abyssal biodiversity. All legal/contractual statuses are subject to change as exploration contracts expire (2026 onwards) and new exploitation regulations are adopted by the ISA.

Sources: See text for key references. Notable sources include ISA documentation and statistics 【63†L213-L221】【46†L1175-L1178】, industry technical reports【43†L1119-L1127】【22†L1130-L1138】, and recent peer-reviewed reviews【6†L47-L54】【62†L25-L31】. All data gaps and estimates are explicitly noted above.
